Network Science

Lecture 06: Structural Balance

Prof. Dr. Michael Granitzer

Dr. Kanishka Ghosh Dastidar

Dr. Jelena Mitrovic

Speaker notes

This lecture moves from unsigned to **signed** graphs — edges carry a positive or negative label representing alliance/rivalry, trust/distrust, like/dislike. The central insight is that a simple local constraint on triangles forces a surprisingly rigid global structure. That interplay between local psychology and global organization is what makes structural balance one of the most elegant results in network science. The lecture continues the course arc by introducing a fourth kind of gap: the Balance Theorem applies to complete graphs, but real networks are sparse and noisy — so we need approximate balance measures, the same modeling/measurement pattern from L03–L05.

From Unsigned to Signed Graphs

So far, an edge meant "connected." Starting now, every edge also carries a **sign**: positive (+) for alliance, friendship, or trust; negative (−) for rivalry, hostility, or distrust.

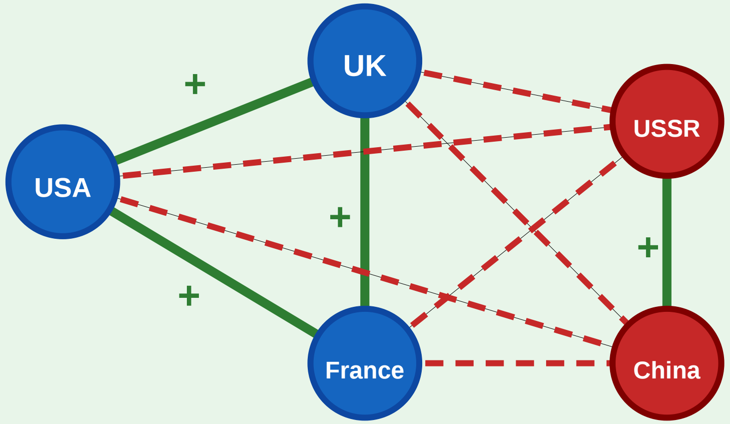
This one addition — a single bit per edge — turns out to constrain global structure powerfully.

A Motivating Scenario

Cold War alliances as a signed graph

green solid = alliance (+) · red dashed = rivalry (−)

Pre-1960 — two balanced blocs



Western Bloc

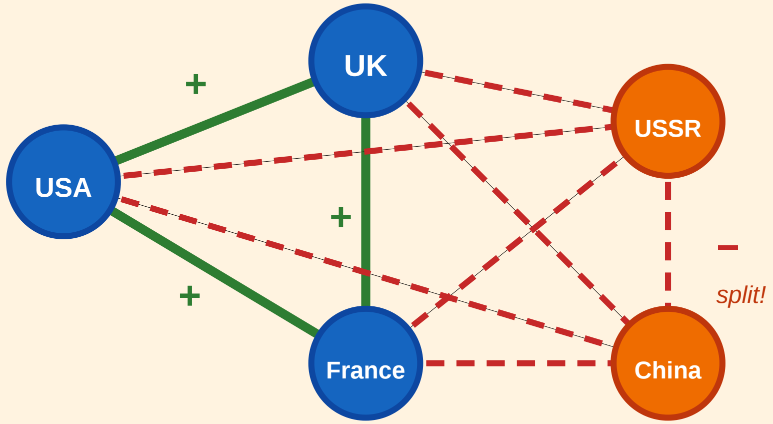
Eastern Bloc

All 10 triangles are balanced

→ Balance Theorem: exactly 2 camps

✓ *strong structural balance*

Post-1969 — Sino-Soviet split



△ USA–USSR–China: all negative (−−−)

→ violates strong balance

→ but satisfies weak balance (≥3 camps)

camps: {USA,UK,FR} · {USSR} · {China}

Five nations during the Cold War. **Left:** pre-1960, all alliances and rivalries form two clean blocs — every triangle is balanced. **Right:** after the 1969 Sino-Soviet split, USSR–China flips to −, creating all-negative triangles. The two-bloc structure breaks.

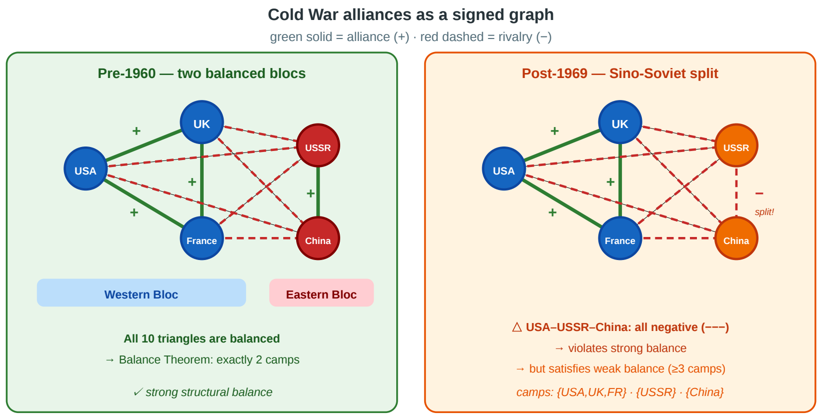
Speaker notes

This figure is the anchor for the lecture. Use it to seed the four questions on the next slide. By the end of the session students should be able to (a) classify every triangle in both panels, (b) explain why the left panel is balanced and the right is not under strong balance, (c) show that the right panel satisfies weak balance with three coalitions, and (d) connect the triangle counts to Leskovec et al.'s empirical findings.

Four Questions from One Picture

1. **Local stability.** Which signed triangles feel psychologically "stable" — and which feel like they should change?
2. **Global structure.** If every triangle is stable, what does that force the *whole* graph to look like?
3. **Relaxation.** When a three-way rivalry (all −) appears, does the two-bloc theorem still hold — or do we need more camps?
4. **Measurement.** In a real signed network with millions of edges, how close to balanced is it — and how would we know?

Every concept in today's lecture answers one of these.



Speaker notes

Questions 1–2 are Module 1 (Balance Theorem). Question 3 is Module 2 (Weak Balance). Question 4 spans Modules 3–4 (empirical test and approximate balance). Keep referring back to the Cold War figure throughout.

A New Kind of Gap

L03–L04 had a **computational** gap: MaxSTC and modularity maximization are NP-hard. L05 had a **causal** gap: cross-sectional data cannot identify selection vs. socialization.

L06 has a **structural** gap: the Balance Theorem requires a *complete* signed graph, but real networks are *sparse*. Most pairs have no observed edge, so the theorem's premise is never met exactly.

Lecture	Gap type	Ideal	Reality	Strategy
L03–L04	Computational	Exact optimization	NP-hard	Polynomial heuristics
L05	Causal	Mechanism identification	Snapshots insufficient	Longitudinal data
L06	Structural	Complete signed graph	Sparse, noisy data	Approximate balance measures

Speaker notes

Naming the gap type up front gives students a framework for understanding why the clean theorems of Module 1 do not directly apply to real data, and why Modules 3–4 are necessary.

Takeaway

> Adding a sign to each edge creates a surprisingly powerful local constraint. The structural balance theorem turns triangle-level psychology into a global two-bloc prediction — but applying it to real data requires dealing with incompleteness and approximation.

Signed Graphs and Structural Balance

> **Guiding Question:** If every triangle in a signed graph is "psychologically stable," what does that force the global structure to be?

Speaker notes

This module introduces signed graphs formally, presents Heider's psychological motivation, classifies the four triangle types, states the Balance Theorem, and sketches its proof. It answers Questions 1 and 2 from the opening.

Heider's Insight (1946)

Fritz Heider (1946), *Attitudes and Cognitive Organization*, Journal of Psychology 21, 107–112.

Heider proposed that people experience psychological tension ("cognitive dissonance") when their relationships form certain patterns — for example, two friends who both dislike the same person feel stable, but two friends who disagree about a third person feel pressure to change.

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Speaker notes

Heider's paper is the origin of the entire field. His psychological intuition — that certain configurations of friend/enemy relationships are stable and others are unstable — was formalized into a graph-theoretic result by Cartwright & Harary (1956). The result is one of the earliest examples of a local psychological rule producing a provable global structural constraint.

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The network-science formalization: encode the tension as a constraint on **signed triangles**.

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Signed Graphs — Formal Definition

> A **signed graph** is a pair (G, σ) where $G = (V, E)$ is a graph and $\sigma : E \to +, -$ assigns a sign to every edge.
>
> - $\sigma(e) = +$: alliance, friendship, trust
> - $\sigma(e) = -$: rivalry, hostility, distrust

A **complete signed graph** has an edge (with some sign) between every pair of nodes.

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Speaker notes

The completeness assumption matters: the Balance Theorem requires every pair to have a signed relationship. In the Cold War scenario, the five nations form a complete signed graph — every pair has a clear alliance or rivalry. In real social networks, most pairs have no observed edge, which is why Module 4 introduces approximate balance.

The Four Signed Triangles

The four signed triangles: +++, ++−, +−−, −−−

A triangle is **balanced** if and only if it has an **even** number of negative edges (0 or 2). The $(+, +, -)$ triangle is unstable: two friends share an enemy, creating pressure for one friendship to break. The $(-, -, -)$ triangle is the ambiguous case — stable under weak balance but not strong.

Speaker notes

The even-number rule is the fastest way to classify triangles. The psychological interpretations are: $(+\,+\,+)$ "all friends" — no tension; $(+\,+\,-)$ "two friends disagree about a third" — tension, someone must change; $(+\,-\,-)$ "the enemy of my friend is my enemy" — stable; $(-!-!-)$ "three-way rivalry" — everyone would benefit from allying with *someone*, but who? The last case is what distinguishes strong from weak balance.

The Structural Balance Theorem

Structural Balance Theorem (Cartwright & Harary, 1956). A complete signed graph (G, σ) is balanced — i.e., every triangle has an even number of negative edges — if and only if the nodes can be partitioned into **at most two** groups such that:

- every edge *within* a group is positive
- every edge *between* groups is negative

In other words: **balance implies two camps**.

Speaker notes

This is one of the cleanest results in network science. A local triangle constraint produces a rigid global structure with only two possible shapes: one all-positive group (if everyone is friends) or two internally friendly groups with mutual hostility. There is no third option. The Cold War pre-1960 panel is the canonical illustration: {USA, UK, France} vs. {USSR, China}.

Proof Sketch

Direction 1 (camps → balance): If the graph splits into ≤2 camps with + inside and − between, then every triangle has 0 or 2 negative edges (check the three possible placements of three nodes across two groups). ✓

Direction 2 (balance → camps): Pick any node v, and put all its friends ($+$ neighbors) in group A with v, and all its enemies ($-$ neighbors) in group B. Now take any two nodes u, w in group A. They are both friends of v, so the triangle v–u–w has two positive edges (v–u and v–w). For the triangle to be balanced, u–w must be $+$. By symmetric reasoning, any two nodes in B are linked by $+$, and any pair $(a \in A, b \in B)$ by $-$. ✓

Speaker notes

The proof of direction 2 is worth walking through explicitly because it shows *why* the local rule cascades globally. Once you fix one node's partition, every other node's assignment is forced. The proof also reveals the key assumption: *completeness*. If some edges are missing, the cascade does not work — which is exactly the structural gap we flagged in the introduction.

Back to the Cold War

In the pre-1960 panel: **all 10 triangles** in K_5 are balanced.

Triangle	Signs	Count of –	Balanced?
USA–UK–France	$+, +, +$	0	✓
USA–UK–USSR	$+, -, -$	2	✓
USA–UK–China	$+, -, -$	2	✓
USA–France–USSR	$+, -, -$	2	✓
USA–France–China	$+, -, -$	2	✓
USA–USSR–China	$-, -, +$	2	✓
UK–France–USSR	$+, -, -$	2	✓
UK–France–China	$+, -, -$	2	✓
UK–USSR–China	$-, -, +$	2	✓

Triangle	Signs	Count of –	Balanced?
France–USSR–China	$-, -, +$	2	✓

The theorem correctly predicts two camps: {USA, UK, France} and {USSR, China}.

Speaker notes

Walking through every triangle on a small example makes the theorem concrete. Students can verify each row, and the pattern becomes visible: the only triangles with 0 negative edges are all-Western; all others have exactly 2 negative edges (one cross-bloc edge per Western pair). The count confirms the prediction mechanically.

Takeaway

> The Structural Balance Theorem is a rare example of a local constraint (triangle psychology) producing a clean global prediction (two camps). A single triangle rule, applied consistently across a complete signed graph, locks the entire network into a rigid two-bloc structure.

Quick Check

Consider four nations with the following signed edges: A–B: +, A–C: +, A–D: −, B–C: +, B–D: −, C–D: −.

1. List all four triangles and classify each as balanced or unbalanced.
2. Is the graph balanced? If so, what are the two camps?
3. If we flip the C–D edge from − to +, which triangle(s) become unbalanced?

Speaker notes

Give them 5 minutes. (1) ABC: +++, balanced. ABD: +−−, balanced. ACD: +−−, balanced. BCD: +−−, balanced. (2) All balanced → yes. Camps: {A,B,C} vs {D}. (3) Flipping C–D to + makes triangle BCD become ++− (unbalanced) and ACD become ++− (unbalanced), while ABC stays +++ and ABD stays +−−.

Quick Check — Solution

1. Triangle classification:

Triangle	Signs	Balanced?
A–B–C	$+, +, +$	✓
A–B–D	$+, -, -$	✓
A–C–D	$+, -, -$	✓
B–C–D	$+, -, -$	✓

2. All balanced → . Camps: A, B, C vs D.

Flipping C–D to + makes:

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Quick Check — Solution

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2. All balanced → **yes**. Camps: A, B, C vs D.

3. Flipping C–D to + makes:

- A–C–D → $+, +, +$ ✓ (still balanced)

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Quick Check — Solution

1. Triangle classification:

Triangle	Signs	Balanced?
A–B–C	$+, +, +$	✓
A–B–D	$+, -, -$	✓
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B–C–D	$+, -, -$	✓

2. All balanced → **yes**. Camps: A, B, C vs D.

3. Flipping C–D to + makes:

- A–C–D → $+, +, +$ ✓ (still balanced)
- B–C–D → $+, +, -$ ✗ (unbalanced — B and C are friends, C and D are now friends, but B and D are enemies)

Weak Structural Balance

Guiding Question: What happens to the two-camps theorem when three-way rivalries are allowed?

Speaker notes

The Sino-Soviet split in the Cold War scenario motivates this module directly. After 1969, the triangle USA–USSR–China is all-negative ($---$). Under strong balance this is forbidden; under weak balance it is allowed. The global consequence: instead of exactly two camps, we get *multiple* camps.

The Problem with All-Negative Triangles

In the post-1969 Cold War panel, the triangle USA–USSR–China has three negative edges.

Under strong balance, this is **unbalanced** — the theorem says the graph cannot split into two camps. But historically, the world *did* function with three separate power blocs: {USA, UK, France}, {USSR}, {China}.

Speaker notes

The psychological distinction is important: $(+,+,-)$ creates active dissonance within the triangle (your friend's enemy is your friend), while $(-,-,-)$ creates no such contradiction — nobody expects their enemy's enemy to be their friend. Davis (1967) formalized this distinction into weak structural balance.

The Problem with All-Negative Triangles

In the post-1969 Cold War panel, the triangle USA–USSR–China has three negative edges.

Under strong balance, this is **unbalanced** — the theorem says the graph cannot split into two camps. But historically, the world *did* function with three separate power blocs: {USA, UK, France}, {USSR}, {China}.

The issue is psychological: a three-way rivalry may be *unstable* (each party would benefit from allying with someone), but it is not *contradictory* in the way that $(+, +, -)$ is. Two friends disagreeing about a third face direct tension; three mutual enemies merely face *opportunity*.

Weak Balance — Definition

Weak Structural Balance (Davis, 1967). A complete signed graph is **weakly balanced** if it contains no triangle with exactly **one** negative edge $(+, +, -)$. All-negative triangles $(-, -, -)$ are permitted.

Speaker notes

The relaxation is small — permit one additional triangle type — but the global consequence is large: instead of ≤2 camps, we can have *any number* of internally friendly, mutually hostile coalitions. The Cold War after 1969 illustrates $k = 3$: Western Bloc, Soviet Bloc, China. This is a powerful example of how changing one local rule changes the entire global picture.

Weak Balance — Definition

> **Weak Structural Balance (Davis, 1967).** A complete signed graph is **weakly balanced** if it contains no triangle with exactly **one** negative edge $(+, +, -)$. All-negative triangles $(-, -, -)$ are permitted.

Weak Balance Theorem. A complete signed graph is weakly balanced if and only if the nodes can be partitioned into $k \geq 1$ groups such that:

- every edge *within* a group is positive
- every edge *between* groups is negative

(Strong balance is the special case $k \leq 2$.)

Back to the Cold War — Post-1969

After the Sino-Soviet split, the five-nation graph has:

- $(+, +, +)$ triangles: USA–UK–France ✓
- $(+, -, -)$ triangles: all Western-pair + one Eastern node (6 triangles) ✓
- $(-, -, -)$ triangles: USA–USSR–China, UK–USSR–China, France–USSR–China ✓ under weak balance

Speaker notes

The check is now easy: students just need to confirm no $(+, +, -)$ triangle exists. The three all-negative triangles are exactly the ones that strong balance forbids but weak balance permits. The result: three camps instead of two, matching the historical reality of a multipolar Cold War after 1969.

Back to the Cold War — Post-1969

After the Sino-Soviet split, the five-nation graph has:

- $(+,+,+)$ triangles: USA–UK–France ✓
- $(+,-,-)$ triangles: all Western-pair + one Eastern node (6 triangles) ✓
- $(-,-,-)$ triangles: USA–USSR–China, UK–USSR–China, France–USSR–China ✓ under weak balance

No $(+,+,-)$ triangles exist → **weakly balanced**.

Partition into 3 camps: $\text{USA, UK, France}, \text{USSR}, \text{China}$.

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Takeaway

Weak balance replaces the two-bloc theorem with a k-coalition theorem. Permitting all-negative triangles allows multiple hostile camps — a better model for real-world multipolar structures like the post-1969 Cold War.

Quick Check

A social network of 6 people has the following signed edges (complete graph):

- Within A, B: all +
- Within C, D: all +
- Within E, F: all +
- Between any two groups: all −

1. Is this graph balanced (strong)? Why or why not?
2. Is it weakly balanced?
3. What is the minimum number of edge flips needed to make it strongly balanced?

Speaker notes

Give them 4 minutes. (1) Not strongly balanced: triangles like A–C–E are all-negative (−−−). (2) Yes, weakly balanced: no $(+,+,-)$ triangles exist. Three camps: {A,B}, {C,D}, {E,F}. (3) To make it strongly balanced (≤2 camps), we need to merge at least two camps. Merging {C,D} into {A,B} means flipping 4 inter-group edges from − to +. So minimum 4 flips.

Quick Check — Solution

Triangles like A–C–E have three negative edges — forbidden under strong balance.

No $(+, +, -)$ triangle exists. Three camps: A, B, C, D, E, F.

Strong balance requires ≤2 camps. To merge {C,D} into {A,B}, flip the 4 cross-group edges (A–C, A–D, B–C, B–D) from − to +. Then the two camps are {A,B,C,D} and {E,F}.

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3. Minimum 4 flips. Strong balance requires ≤2 camps. To merge {C,D} into {A,B}, flip the 4 cross-group edges (A–C, A–D, B–C, B–D) from − to +. Then the two camps are {A,B,C,D} and {E,F}.

Empirical Test: Signed Networks in Social Media

Guiding Question: In real signed networks with millions of edges, does balance actually hold?

Speaker notes

This module provides the empirical anchor for L06, following the same pattern as Kossinets-Watts (L03), Zachary (L04), and Christakis-Fowler (L05). The study is Leskovec, Huttenlocher & Kleinberg (2010), which analyzed three large online signed networks and found that triangle-level balance holds remarkably well — but with important caveats.

The Study

Leskovec, Huttenlocher & Kleinberg (2010), *Predicting Positive and Negative Links in Online Social Networks*, Proc. 19th International Conference on World Wide Web (WWW), 641–650.

They analyzed three large signed networks:

Platform	Nodes	Signed edges	Type
Epinions	131 828	841 372	trust / distrust
Slashdot	82 144	549 202	friend / foe
Wikipedia	7 118	103 747	support / oppose (admin elections)

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Speaker notes

These are among the largest signed networks ever studied. Epinions was a product-review platform where users could mark others as trusted or distrusted. Slashdot let users tag others as "friend" or "foe." Wikipedia admin elections record public support/oppose votes. All three provide explicit positive and negative edges — no inference needed.

What They Found

Triangle-level balance is remarkably strong:

Triangle type	Expected (random signs)	Observed (Epinions)
$+, +, +$	~12.5%	~47%
$+, +, -$	~37.5%	**~8%**
$+, -, -$	~37.5%	~37%
$-, -, -$	~12.5%	~8%

Speaker notes

The random baseline assumes each edge is independently + or − with the observed overall positive rate. The deviation from this baseline is the evidence for balance. The key number: $(+, +, -)$ is suppressed by a factor of ~5×. This is the strongest possible evidence that local balance constraints shape global structure in real signed networks. The $(-, -, -)$ underrepresentation is also noteworthy — it falls between the strong and weak balance predictions.

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The $(+, +, -)$ type — the one both strong and weak balance forbid — is **massively underrepresented** (~8% vs. 37.5% expected). The balanced types $(+, +, +)$ and $(+, -, -)$ are overrepresented. Balance is not perfect, but the skew is dramatic.

What the Study Could Not See

1. Sparse, not complete. The Balance Theorem assumes a complete signed graph. These networks are extremely sparse — most pairs have no edge. The theorem's prediction (two or k camps) does not directly apply; the study tests triangle statistics, not the global partition.

2. Directed signs. On Epinions, trust and distrust are *directed* — A can trust B without B trusting A. The balance framework assumes undirected signs. Directed balance theory exists (Leskovec et al. explore it) but is less clean.

3. Missing negative edges. People are less likely to *create* a negative edge than to simply *not interact*. The observed negative edges are a biased sample: they represent actively expressed hostility, not indifference. The "true" signed graph — if every pair had a relationship — might look different.

4. Temporal dynamics. The data is a snapshot. Balance theory predicts an *equilibrium*; the snapshot might capture a network mid-transition, with unbalanced triangles that are in the process of resolving.

Speaker notes

This follows the L03–L05 pattern: celebrate the result, then identify blind spots. The sparsity issue is the most fundamental: the Balance Theorem is a statement about complete graphs, and no real network is complete. The study's response — test triangle statistics rather than global partition — is a pragmatic substitute, but it leaves open whether the *global* structure is consistent with balance. Module 4 introduces tools for addressing this gap.

Takeaway

> Leskovec et al. (2010) showed that balance holds at the triangle level in large online signed networks: the forbidden $(+, +, -)$ type is massively underrepresented. But real networks are sparse and directed, so the clean two-camp prediction of the Balance Theorem must be tested through approximate measures, not exact partition.

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Incomplete and Approximate Balance

Guiding Question: How can we assess balance when the graph is incomplete or only approximately balanced?

Speaker notes

This module addresses the structural gap identified in the introduction: the Balance Theorem requires completeness, but real networks are sparse. We need tools that work on partial data. Two approaches: the cycle-based criterion (extend balance from triangles to all cycles) and the frustration index (measure distance from the nearest balanced graph).

From Triangles to Cycles

On a complete graph, checking triangles is sufficient for balance (the theorem guarantees it). On an *incomplete* graph, triangles alone are not enough — balance must be checked on **all cycles**.

> **Cycle criterion.** A signed graph (G, σ) is balanced if and only if every cycle in G has an **even** number of negative edges.
>
> Equivalently: no cycle has an odd number of negative edges.

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Speaker notes

This is Harary's generalization (1953). On a complete graph it reduces to checking triangles, because every longer cycle can be decomposed into triangles and the triangle condition implies the cycle condition. On a sparse graph, the cycle condition is strictly stronger: a graph can have all balanced triangles but still contain unbalanced longer cycles. The cycle criterion is the formally correct definition for incomplete signed graphs.

The Frustration Index

In real data, perfect balance is rare. The natural question becomes: **how far** is the graph from balanced?

Frustration index $F(G, \sigma)$: the minimum number of edges whose sign must be flipped to make the graph balanced.

$$F(G, \sigma) = \min_{\sigma'} |e \in E : \sigma(e) \neq \sigma'(e)|$$

subject to (G, σ') being balanced.

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Speaker notes

The frustration index is the signed-graph analogue of the edit distance to the nearest balanced partition. Computing it exactly is NP-hard in general (it reduces to a graph-cut problem), but good approximations exist — including spectral methods that use the signed Laplacian. The normalized version f provides a scale-independent measure that can be compared across networks.

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subject to (G, σ') being balanced.

Normalized frustration: $f = F/|E|$. If $f \approx 0$, the graph is near-balanced; if $f \approx 0.5$, it is no more balanced than a randomly signed graph.

Computing Approximate Balance

Exact computation. Finding F exactly is NP-hard (equivalent to a minimum-weight graph cut on a related unsigned graph). On small graphs, integer programming or brute force works; on large graphs, we need heuristics.

Spectral approach. The **signed Laplacian** $L_\sigma = D - A_\sigma$ (where A_σ has entries $+1$ or -1) has the property that (G, σ) is balanced iff the smallest eigenvalue $\lambda_1(L_\sigma) = 0$. The magnitude of λ_1 quantifies how far the graph is from balanced.

Practical rule of thumb. For large networks, count the fraction of balanced triangles ($T_{\text{bal}}/T_{\text{total}}$) as a fast proxy. Leskovec et al.'s Epinions data had $T_{\text{bal}}/T_{\text{total}} \approx 0.92$.

Speaker notes

The spectral approach connects to L04's eigenvector centrality: both use properties of the graph's matrix representation. The signed Laplacian is the natural extension of the unsigned Laplacian to signed graphs, and its spectrum encodes balance information just as the unsigned Laplacian's spectrum encodes connectivity information. For large networks, the balanced-triangle fraction is the cheapest informative statistic — it is what Leskovec et al. actually computed.

Quick Check

A signed graph on 5 nodes has 7 edges. You compute that 8 out of 10 triangles are balanced and 2 are unbalanced (both of type $+, +, -$).

1. Is the graph balanced? Why or why not?
2. Can you conclude that the frustration index is exactly 2?
3. What additional information would you need to determine the exact frustration index?

Speaker notes

Give them 4 minutes. (1) No — balanced requires *all* triangles (and all cycles) to be balanced. Two unbalanced triangles means the graph is not balanced. (2) No — the frustration index counts *edge flips*, not unbalanced triangles. Two unbalanced triangles might share an edge; flipping that one edge could fix both. So $F \leq 2$ but could be 1. (3) You would need the full edge list and sign assignment to determine which specific edges, when flipped, resolve all imbalances with the fewest changes.

Quick Check — Solution

Two unbalanced triangles violate the "every triangle must be balanced" condition.

2. No. Frustration index counts , not unbalanced triangles. The two bad triangles might share an edge — flipping that one shared edge could fix both. So $F \leq 2$, but F could be as low as 1.

— which nodes and edges form the unbalanced triangles, and whether they share edges. Determining F exactly requires solving a minimum-cut problem on the graph (NP-hard in general, but tractable on small instances).

Quick Check — Solution

1. Not balanced. Two unbalanced triangles violate the "every triangle must be balanced" condition.

2. No. Frustration index counts , not unbalanced triangles. The two bad triangles might share an edge — flipping that one shared edge could fix both. So $F \leq 2$, but F could be as low as 1.

— which nodes and edges form the unbalanced triangles, and whether they share edges. Determining F exactly requires solving a minimum-cut problem on the graph (NP-hard in general, but tractable on small instances).

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3. You need the full graph — which nodes and edges form the unbalanced triangles, and whether they share edges. Determining F exactly requires solving a minimum-cut problem on the graph (NP-hard in general, but tractable on small instances).

Takeaway

Real signed networks are sparse and imperfect. The frustration index and the balanced-triangle fraction are the practical tools for measuring how close a network is to the balanced ideal — replacing the all-or-nothing theorem with a quantitative diagnostic.

Wrap-Up — The Four Gaps

Across Lectures 03–06, the course has exposed four kinds of gap between theory and data:

Lecture	Gap type	Theoretical ideal	Practical limitation	Strategy
L03	Computational	True tie labels (MaxSTC)	NP-hard	Polynomial proxies
L04	Computational	Optimal partition ($\max Q$)	NP-hard	GN, Louvain, Leiden
L05	Causal	Mechanism identification	Snapshots insufficient	Longitudinal data
L06	Structural	Complete balanced graph	Sparse, noisy data	Frustration index, spectral methods

Every concept in the four lectures occupies one cell of this table.

Looking Ahead: Lecture 07

How can a network be both locally clustered *and* globally navigable? Small-world networks show that a handful of random long-range shortcuts dramatically reduce path lengths — reconciling clustering and short paths without sacrificing either.

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Reading

Chapter 5 of Easley & Kleinberg (2010) covers structural balance theory, weak balance, and applications. Heider (1946) and Cartwright & Harary (1956) are the original sources. Leskovec, Huttenlocher & Kleinberg (2010) is the empirical study on signed social networks.

Speaker notes

The four-gap table closes the L03–L06 arc. Each lecture introduced a different kind of obstacle between the theoretical construct and the real-world measurement: computational (NP-hardness), causal (unidentifiable mechanisms), and structural (completeness assumptions unmet). All four share the same response pattern: fall back on approximate tools and be explicit about what the approximation costs. Lecture 07 changes direction: instead of analyzing existing networks, it asks how networks can *simultaneously* exhibit two seemingly contradictory properties — high clustering and short paths — and introduces generative models (Watts–Strogatz) for the first time.